

CSE 321 Quiz 04 Set A

Time 25 Minutes

Name	ID	Sec
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Q1. In a system there are 3 processes- **P1 (9 bytes)**, **P2 (6 bytes)** and **P3 (7 bytes)** with page size of 2 bytes. The size of the main memory is 32 bytes. Page tables of processes are given below.

Page Tables					
Process P1		Process P2		Process P3	
p	f	p	f	p	f
0	7	0	15	0	2
1	12	1	13	1	6
2	1	2	9	2	3
3	5			3	8
4	4				

Find corresponding physical addresses of the following logical addresses:

[1+1+1]

- Address 01001 of P1
- Address 00100 of P1.
- Address 11011 of P3.

Q2. Consider this Page table where each page = 4 Byte. Virtual Address Space 7 bits. answer the following questions:

[1+1]

Page	Frame	Validity
0	7	v
1		i
2		i
3	9	v

- In the Backing Store, how much memory/space will it take to make the page table information consistent.
- CPU asked for this logical address: $(15)^{10}$ (15 decimal). Give me the Actual Physical Location of this virtual address

Q3. There are a total of 3 frames in your main memory. And your page table has a total of 5 frames. You need to accommodate them. Using FIFO and LRU page replacement techniques, Who will give the higher Page Hit Ratio?

[2.5 + 2.5]

Reference String:

[0 1 2 1 3 1 0 4 5 1 3 4]